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RECEIVED 13 December 2025

REVISED 28 April 2026

ACCEPTED 29 June 2026

PUBLISHED 17 July 2026

CITATION

Zhou X and Liu S (2026) Digital
infrastructure construction and industrial
chain resilience: empirical evidence
from high-tech industry in China.
Front. Sustain. 7:1767126.
doi: 10.3389/frsus.2026.1767126

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Digital infrastructure construction and industrial chain resilience: empirical evidence from high-tech industry in China

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Introduction: Currently, China's high-tech industrial chain faces a "dual challenge" both internally and externally. Enhancing high-tech industrial chain resilience (HCR) has become a critical issue for safeguarding China's industrial security. As a crucial foundation for the digital era, digital infrastructure construction (DIC) provides significant support for improving HCR.

Methods: Based on panel data from 31 regions in China from 2008 to 2021, this paper employs panel regression model and mediation effect model to examine the mechanisms through which DIC influences HCR.

Results: The empirical results indicate that DIC has a significant positive impact on HCR, with an average effect of 7.2%. This impact is mediated through three pathways: industrial co-agglomeration, innovation efficiency, and factor allocation level. The heterogeneity tests reveal that the promoting effect is more pronounced in the eastern and southern regions, while it is relatively weaker or even negative in the central, western, and northern regions.

Discussion: Overall, future efforts should focus on enhancing the network correlation between the hightech industry and producer services through the rational layout of DIC, supporting the construction of innovation institution and platform environment for high-tech industrial chain with a "function-oriented" approach, and improving factor allocation level by refining the regulatory mechanisms for "labor-capitaldata." This will help foster a more cohesive, dynamic, resilient, and competitive high-tech industrial chain.

KEYWORDS

digital infrastructure construction, high-tech industry, industrial chain resilience, mediation effect model, network externalities

1 Introduction

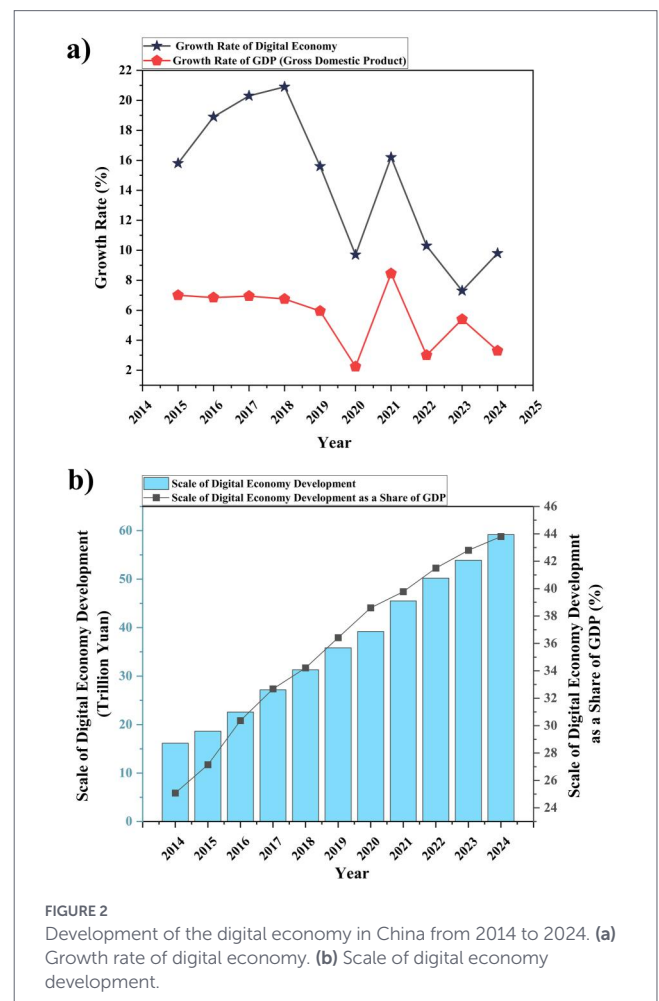
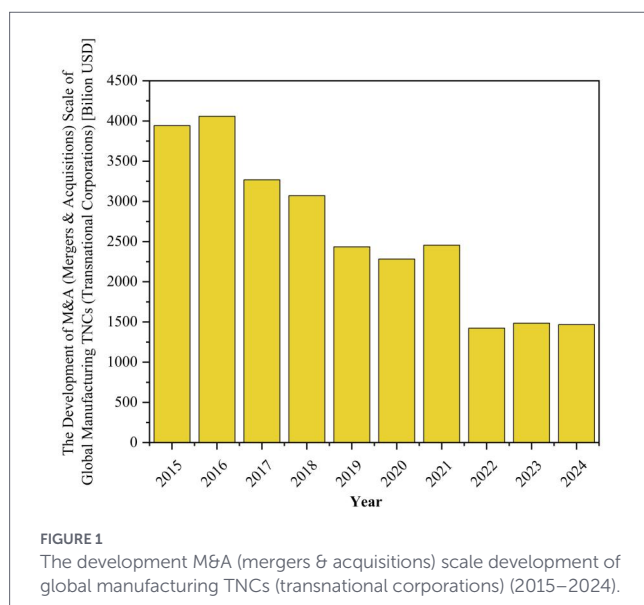
The ongoing wave of technological revolution and industrial transformation is profoundly reshaping modern industrial development, which increasingly emphasizes regional division of labor, trade liberalization, and scientific and technological support. This has ushered in the era of "chain clusters." However, in recent years, as both internal and external situations have become increasingly severe, China's industrial chain development faces a "dual challenge." Externally, factors such as the Trade disputes between China and the United States, the COVID-19 pandemic, and geopolitical conflicts have introduced numerous global uncertainties that threaten industrial chain security (Du et al., 2023). The global manufacturing sector is increasingly shifting toward localization and shorter supply chains. According

to the UNCTAD (UN Trade and Development) Cross-Border M&A (Mergers & Acquisitions) Database, global manufacturing multinational corporations have become more cautious in their investments, with the scale of mergers and acquisitions dropping from 394.208 billion dollars in 2015 to 146.737 billion dollars in 2024, a decline of 62.78% (Figure 1). This profoundly reflects the transformation of the underlying logic of the global manufacturing landscape, where the traditional integrated model of “*global production-global manufacturing*” is gradually withdrawing from the historical stage. Under current political rules and cost constraints, enterprises are compelled to choose risk mitigation and gradual reshoring to achieve maximum control over their industrial chain, thereby building localized technological moats. Internally, while China’s industrial chains occupy a high position in the global value chain in terms of “quantity,” they still lag behind developed countries in terms of “quality,” with key core technologies facing “bottleneck” issues. According to a 2018 Ministry of Industry and Information Technology (MIIT) study of more than 130 key basic materials from more than 30 large enterprises in China, the results show that 32% of key materials are still unavailable domestically, and 52% rely on imports. Additionally, over 95% of high-end chips for computers and servers and more than 70% of processors for smart devices are imported. In October 2022, the 20th National Congress of the Communist Party of China emphasized the need to “enhance the resilience and security of industrial and supply chains,” making the strengthening of industrial chain resilience a key issue for China’s high-quality economic development.

On the other hand, the widespread application of new-generation digital technologies such as artificial intelligence, big data, the Internet of Things (IoT), and 5G communication has driven the rapid development of the digital economy. According to the China Digital Economy Development Research Report (2025) released by the China Academy of Information and Communications Technology in April 2026, China’s scale of digital economy development reached 59.2 trillion yuan in 2024, accounting for 43.8% of GDP (Gross Domestic Product), far exceeding China’s GDP growth rate (Figure 2). The digital economy has quietly emerged as an accelerator of national economic growth, profoundly reshaping the traditional

economic driving structure of “investment-consumption-trade.” Through its deep integration with the real economy, it has generated systemic “digital dividends” of cost reduction and efficiency enhancement. As a critical foundation for the digital economy development, DIC provides essential support for the digital economy, playing a pivotal role in enabling the integration of the digital economy with the real economy. According to data from China Academy of Information and Communications Technology, the length of optical cable lines in China has increased to 6,432 kilometers, a 3.68-fold increase from 2013. The number of 5G base stations has reached 3.377 million, and the national computing power hub nodes under the “East Data, West Computing” project are steadily advancing, with the core computing power industry exceeding 1.8 trillion yuan. As DIC continues to expand and improve, its connection with socio-economic development has become increasingly close, gradually integrating into various segments of the industrial chain and value chain. This has provided significant “digital dividends” for China’s technological system innovation and enterprise reform, while also permeating government, public, and social governance reforms (Liu et al., 2020). It has effectively unleashed economic growth potential and facilitated the transition from old to new growth drivers.

High-tech industries, characterized by high added value, high returns, low energy consumption, and low pollution, have always been catalysts for rapid regional economic development. They are also powerful pillars for improving economic structure, enhancing national technological development levels, and boosting



comprehensive competitiveness (Ren and Tang, 2024). According to the China High-tech Industry Yearbook and the World Bank database, it shows that as of 2022, the number of enterprises in China's high-tech industry exceeded 50,000, the value added of high-tech industry above designated size increased by 7.4% compared with the previous year, and the actual use of foreign capital was 444.9 billion yuan with an overall positive development trend (Figure 3). However, compared with the developed countries, there is still a big gap that some of the core indicators are ranked in the lower middle class. The pace of development of high-tech industry is still relatively slow and characterized by the predicament of “low-end locking.” The essence of this predicament lies in the fact that the value distribution of China's high-tech industry is still in a stage of dependency. On the one hand, it is constrained to the low end of the value chain by technological bottlenecks such as intellectual property barriers and export controls on critical equipment. On the other hand, the domestic high-tech industry is generally characterized as being large but not strong, with enterprises lacking sufficient “technological catch-up” capabilities (Zhu and Li, 2025). Against the backdrop of the increasingly prominent strategic position of DIC, how to leverage its power to build more cohesive, dynamic, resilient, and competitive high-tech industrial chain has become a focal issue for ensuring the

security and stability of China's industrial and supply chains. Therefore, this paper focuses on the two core subjects of DIC and HCR, aiming to reveal the underlying mechanisms and propose practical policy recommendations for enhancing the resilience of China's high-tech industrial chain.

The remainder of the paper is organized as follows: Section 2 provides a literature review. Section 3 presents the research hypotheses. Section 4 discusses the research methodology and data sources. Section 5 presents the empirical results, including benchmark regression, instrumental variable tests, robustness checks, mediation effect tests, and heterogeneity tests. Section 6 concludes with policy implications. The potential contributions of this paper are as follows: (1) In terms of research perspective, this paper shifts the focus from “industry” to “industrial chain” and introduces network externality theory, conceptualizing the industrial chain as a dynamic network system composed of multiple nodes. This approach enriches the theoretical and empirical research on industrial chain resilience from both macro and micro perspectives, providing a foundation for subsequent studies. (2) In terms of research content, this paper innovatively combines DIC with industrial chain resilience, focusing on the high-growth high-tech industry. This broadens new pathways for incremental development and contributes to the goal of high-quality economic development in China.

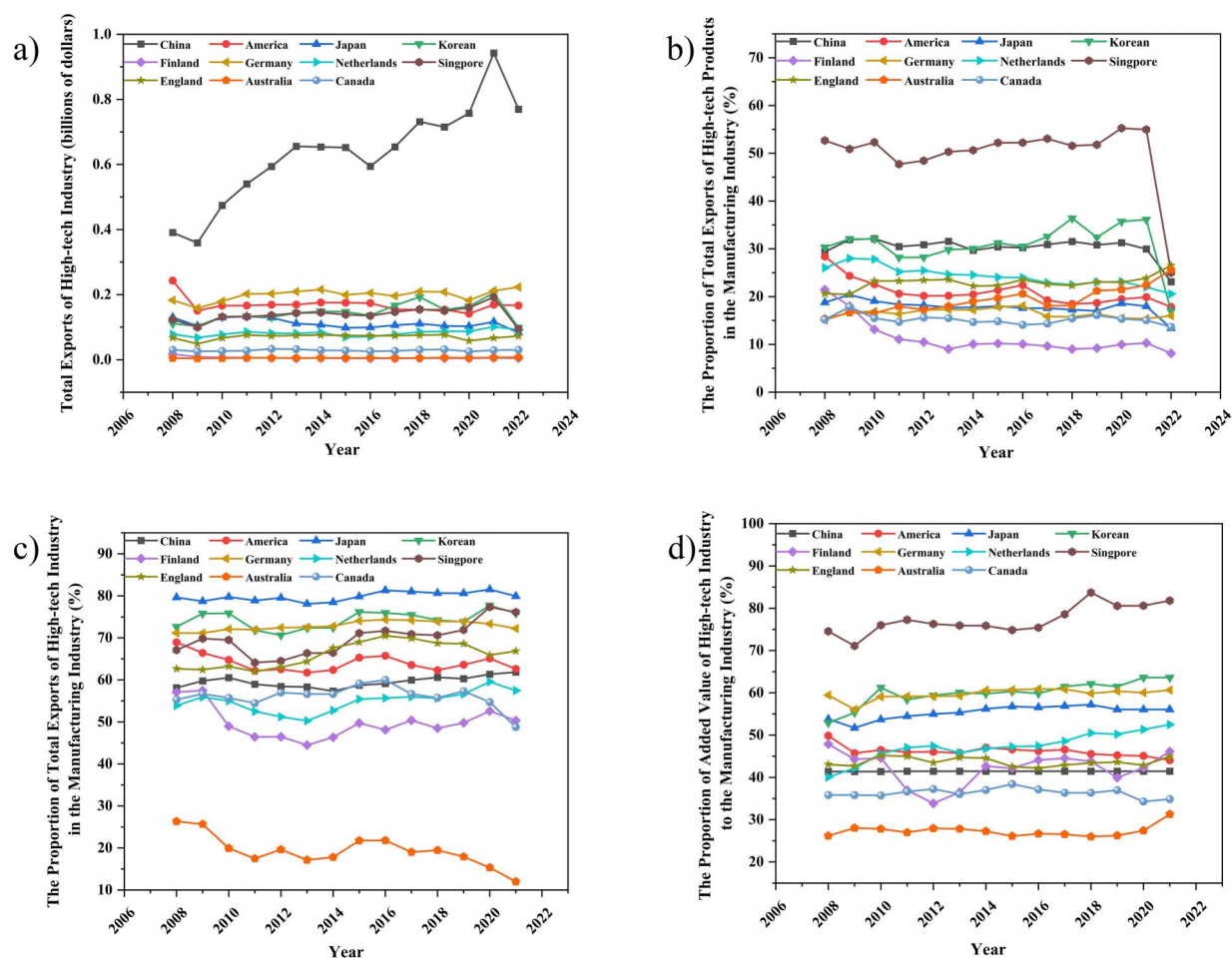


FIGURE 3

Development of high-tech industry in (selected) countries of the world. (a) Total exports of high-tech industry from 2008 to 2022. (b) The proportion of total exports of high-tech products in the manufacturing industry from 2008 to 2022. (c) The proportion of total exports of high-tech industry in the manufacturing industry from 2008 to 2021. (d) The proportion of added value of high-tech industry to the manufacturing industry from 2008 to 2021.

2 Literature review

2.1 Research on the economic effects of DIC

In essence, DIC is a comprehensive infrastructure system. It relies on core information technologies such as artificial intelligence, blockchain, and the industrial internet, and is supported by data and computing power. This system provides services like digital transformation, intelligent upgrading, and integrated innovation, and is applied across all socio-economic fields. DIC breaks traditional geographical barriers, accelerates the advancement of information technology and the rapid flow of knowledge across regions, and provides a continuous source of momentum for China's economic development. Research on its economic effects has gradually become a hot topic in academia. From a macro perspective, DIC has a significant positive impact on regional coordinated development (Li et al., 2024), industrial structure upgrading (Lei et al., 2025), foreign trade exports (Zhou et al., 2022), labor employment rates (Zhou et al., 2024), and factor allocation efficiency (Pan and Yang, 2024). From a micro perspective, DIC positively influences innovation capabilities (Tian and Lu, 2023), entrepreneurial activity (Schade and Schuhmacher, 2022), enterprise performance growth (Jia and Li, 2025), and labor income (He et al., 2026). Many scholars have used mediation effect models and difference-in-differences models to verify these effects. On the other hand, the rapid construction of digital infrastructure can also lead to digital divides and socio-spatial inequalities. For example, Zhang and Sun (2023) focused on China's administrative border regions and found that DIC can promote economic convergence between border regions and central regions, alleviating economic disparities. Jiang and Jin (2024) argued that the impact of DIC on the urban-rural income gap exhibits a threshold effect.

2.2 Research on the concept, measurement, and influencing factors of industrial chain resilience

The concept of "industrial chain resilience" should be understood from two dimensions: "industrial chain" and "resilience." The "industrial chain" originates from the idea of the division of labor by Adam Smith. It is a chain-like, network-based industrial organization where enterprises serve as nodes and coordinated economic activities act as the connecting medium. "Resilience" has generally undergone an iterative evolution through "*engineering resilience* - *ecological resilience* - *evolutionary resilience*." This progression moves from the physics-based ability to "return to the original state," to the capacity to recover and reorganize under change (Holling, 1973), and finally to the ability to withstand external shocks (Reggiani et al., 2002; Walker et al., 2004). Simmie and Martin (2010) summarized this through a theoretical framework of "*reorganization* - *development* - *maintenance* - *release*." It represents a dynamic equilibrium of adaptation and adjustment capabilities, primarily encompassing resistance, recovery, adaptation, and renewal (Martin and Sunley, 2015). Unlike "risk" resistance, which aims to maintain a system's long-term internal stability, "resilience" places greater emphasis on short-term response speed and the capacity for self-renewal and recovery. Therefore, industrial chain resilience refers to the capabilities of nodal enterprises and their

inter-firm relationships within the chain to resist, recover, and recreate after systemic risks.

The measurement of industrial chain resilience mostly draws on methods for measuring economic resilience, which can be broadly divided into four categories: First, core variable measurement methods. Chen and Wen (2023) used network topology core indicators to express the comprehensive ability of industrial chains to resist and recover from risks. Second, comprehensive index system measurement methods. Wu et al. (2023) used the 2F-PT-TOPSIS model to select 13 indicators to measure the resilience of coal industrial chain and supply chain in Shaanxi Province, China. Le (2023) updated this approach by selecting 23 tertiary indicators from four aspects: risk resistance, organizational adaptability, autonomous control, and leading competitiveness to measure the resilience of provincial industrial chain in China. Third, input-output measurement methods based on production networks. Li et al. (2022) used China's input-output tables and the PR algorithm to calculate the global industrial chain embedding probability under industrial chain division scenarios to reflect the industrial chain resilience. Ma L. et al. (2023) used the Eora MRIO multi-regional input-output table and combined the PR algorithm with a social network model to construct a global index of industrial chain resilience for various countries. Fourth, complex network attribute measurement methods. Crespo et al. (2014) focused on regional industrial knowledge networks and used degree distribution and degree correlation as important network attributes to reflect the industrial chain resilience.

The influencing factors of industrial chain resilience mainly focus on digital transformation, internal industrial structure, and technological innovation capabilities. For example, Huang et al. (2023) first used the fsQCA method to analyze the complex causal relationship between new industrialization and the resilience of the manufacturing industrial chain from a macro perspective, and then used the DEMATEL-ISM-BN model to explain the internal mechanism from a micro perspective. Yang and Liu (2024) used a mediation effect model to find that DIC can enhance the industrial chain resilience by optimizing industrial structure, improving capacity utilization, promoting industrial co-agglomeration, and enhancing regional innovation efficiency. Li et al. (2026) employed a time-varying difference-in-differences model and concluded that digital infrastructure can enhance industrial chain resilience through informatization enhancement and innovation-driven effect.

2.3 Research on the impact of DIC on high-tech industries

Existing research on DIC and high-tech industries primarily focuses on the digital economy as the explanatory variable, examining its impact on the innovation capabilities, innovation performance, and innovation efficiency of high-tech industries. For example, Tan (2024) used a three-stage DEA model to find that the innovation efficiency of high-tech industries is positively correlated with digital economy development: for every 1% increase in the digital economy development index, innovation efficiency increases by 0.2102%. Wan et al. (2023) found that digital technology significantly promotes innovation in China's high-tech industries, but the mediation effects of R&D resource investment and technology diffusion exhibit threshold effects. Liu Y. et al. (2024) constructed a spatial Durbin model and a semi-parametric spatial lag model for empirical analysis, finding that

the partial derivative of the digital economy's impact on high-tech industry innovation shows early fluctuations and later stability. Zhang et al. (2024) used the Hicks-Moorsteen index method to examine the impact of the digital economy on the total factor productivity and its decomposition efficiency in China's high-tech industries. The threshold effect shows that the innovation efficiency of high-tech industries reaches its optimal level when the digital economy composite index is between 2.611 and 2.747.

Although the existing literature has laid a solid foundation for understanding the empowerment mechanisms and effects of DIC on high-tech industries, research analyzing HCR from an industrial chain perspective is still in its infancy, and the relevant empirical evidence remains limited. Amid the current environment of intensifying geopolitical conflicts and successive external uncertainty shocks, the research value of this topic has become increasingly prominent. Grounded in network externality theory, this paper systematically analyzes the relationship between DIC and HCR, thereby contributing to the formulation of more targeted and comprehensive policy paradigms and providing referential insights for the rational spatial layout of DIC and the resilient development of high-tech industrial chain.

3 Research hypotheses

3.1 Analysis of the direct impact mechanism

DIC, as a new type of infrastructure that integrates traditional infrastructure with modern information and digital technologies, is essentially aimed at enabling the efficient flow and value creation of data elements through network connectivity. It realizes the integration of data elements with traditional production factors. Through the penetration and integration of digital and real economies and network interconnection, it strengthens the development capabilities of various nodes in high-tech industrial chain and the connections between them, thereby enhancing HCR. Theoretically, this process can be explained through network externalities. Network externalities refer to the phenomenon whereby the value of a network increases at a rate proportional to the square of the number of network nodes; that is, the application of network products continuously attracts new users, and the addition of new users generates more information, which in turn induces further user growth, thereby realizing demand creation and re-creation (Metcalf, 1995). Furthermore, network externalities can be divided into direct network externalities and indirect network externalities: the former is manifested in the direct enhancement of network value through an increase in the number of users, while the latter is reflected in the expansion of user scale driving the development of complementary factors and supporting services, indirectly enhancing the overall utility of the network.

Specifically, first, DIC unleashes direct network externalities by strengthening network connectivity. The rapid construction of information infrastructure such as broadband networks, 5G, and the Internet eliminates geographical barriers, reduces information acquisition and transaction costs, mitigated transactional friction, simplifies redundant information links, and improves the efficiency of various economic activities. This breaks the "information silo" dilemma between upstream and downstream segments of high-tech industrial chain (Fan et al., 2025), enabling cross-temporal communication between different enterprise entities on the chain. It

ensures the smooth operation of different production and circulation segments of high-tech industrial chain, significantly improving the development efficiency and flexibility of the industrial chain. This allows high-tech industrial chain to respond quickly to internal and external risks and shocks, effectively dispersing shocks and mitigating economic fluctuations, thereby enhancing HCR. Second, according to Metcalfe's Law, DIC combines digital technology, data elements, and invested capital to achieve ubiquitous connectivity and effective integration, thereby amplifying indirect network externalities to accelerate the digital transformation process of high-tech industrial chain (Ma R. et al., 2023). This accelerates the digital transformation of high-tech industrial chain. During this process, decision flows, human flows, logistics, capital flows, and technology flows within the industrial chain gradually achieve digital transmission, accelerating the digital reshaping of various business processes (Ahmad et al., 2021). This ultimately drives the comprehensive digital transformation of high-tech industrial chain toward a complex, multi-dimensional spatial network structure. This effectively enhances the ability of high-tech industrial chain to disperse risk shocks, significantly weakening the transmission of risk shocks along the industrial chain, thereby increasing the stability of high-tech industrial chain. Third, as the network structure becomes increasingly complex, DIC further deepens network functions through the in-depth mining and application of data elements. Leveraging the empowering role of digital technology, the ability of high-tech industrial chain to collect and interpret real-time data is significantly improved. It helps high-tech industrial chain obtain real-time data on production, processing, circulation, consumption, and distribution, such as production and processing data, enterprise operation data, user data, and payment data. It conducts vertical penetration and in-depth analysis of this information, solving the "black box" problem of supply and demand information. Therefore, high-tech industrial chain can autonomously and dynamically adjust its production plans to improve management effectiveness and develop "localized" business models based on user needs (Kangmin et al., 2023), expanding the innovation space of high-tech industrial chain and thereby enhancing its resilience. In summary, under the mechanism of network externalities, DIC, through the progressive process of forging "linkage expansion-synergistic integration-functional reinforcement," continuously empowers the cross-regional operation, risk dispersion, and digital embedding capabilities of high-tech industrial chain, thereby significantly enhancing its resilience. Based on the above analysis, this paper proposes Hypothesis 1:

H1: DIC has a positive impact on HCR.

3.2 Analysis of the indirect impact mechanism

From the perspective of direct network externalities, the preceding analysis has examined the direct facilitating effect brought about by DIC through strengthening network connectivity and reducing information and transaction costs. Furthermore, from the perspective of indirect externalities, network value is enhanced not only through direct connectivity relationships, but is further amplified through the agglomeration of complementary factors, the continuous improvement of supporting service systems, and the deepening and expansion of network functions. Under this theoretical logic, the indirect

network externalities of DIC are primarily manifested in three inter-related dimensions with distinct impact pathways:

- 1 At the level of network structure, the expansion of network value drives the reconfiguration of nodal relationships and the spatial pattern of high-tech industrial chain. DIC expands the advantages of regional digital technology development, effectively reducing the production costs of high-tech industrial chain and the communication and coordination costs between upstream and downstream enterprises, creating economies of scale. According to the center-periphery theory, various production factors from outside the region will quickly flow into regions with technological advantages and economies of scale to access abundant development opportunities and economic resources, continuously promoting the dynamic evolution of spatial agglomeration and diffusion (Zeng et al., 2024). This fosters the continuous agglomeration of high-tech industrial enterprises. On the other hand, DIC investments are more closely related to R&D technologies such as 5G, big data, artificial intelligence, and blockchain, which will generate demand for necessary productive services such as technology R&D, network operations, hardware and software design, industrial data collection-storage-analysis-modeling, smart logistics, and platform operations. This creates more employment opportunities in related productive service industries, covering real estate, finance, transportation, warehousing, postal services, scientific research, technical services, geological exploration, information transmission, computer services, and software (Guo et al., 2020). Thus it promotes the agglomeration of productive service industries. DIC serves as a carrier for the agglomeration and circulation of “data” elements, connecting high-tech industries and productive service industries through digital information-sharing platforms, promoting the co-agglomeration of the two industries. From a production perspective, industrial co-agglomeration provides a favorable external environment for cross-industry knowledge dissemination, improving knowledge transfer efficiency, reducing knowledge dissemination costs, and strengthening knowledge spillover effects (Grieser et al., 2022). It enhances the absorptive capacity of the high-tech chain to external economic fluctuations. Productive service industries can provide more professional and refined services to high-tech industries, promoting the emergence of more new technologies, products, and development models (Chen, 2019), thereby enhancing the path-recreation capability of high-tech industrial chain. From a circulation and consumption perspective, the collaborative exchange between productive service industries and high-tech industries can generate “mutual assistance” effects, enhancing market accessibility and enabling the production of specialized and personalized products based on consumer demand. This provides a continuous source of innovation for high-tech industrial chain (Orenstein, 2021). Based on the above analysis, this paper proposes Hypothesis 2:

H2: DIC fosters industrial co-agglomeration, thereby enhancing HCR.

- 2 At the level of network function, the deep integration of data elements and knowledge factors promotes the efficiency of

information interaction and synergy among nodes. The innovation activities exhibit the characteristic of increasing returns to scale. Innovation capability serves as a core indicator determining the strength of market competitiveness. Therefore, from a more micro perspective, optimizing innovation resource inputs and increasing innovation outputs, thereby improving innovation efficiency, is key to enhancing HCR. Innovation efficiency refers to the ratio of technological innovation input resources to technological innovation output. When input is lower and output is higher, innovation efficiency is higher, and vice versa (Wang et al., 2020). DIC improves innovation efficiency in two main ways: in terms of innovation input, DIC promotes the widespread application of new information technologies such as artificial intelligence and big data, broadening information access channels, improving information accessibility and transparency, and continuously advancing the construction of digital sharing platforms. On the one hand, high-tech enterprises can select more suitable R&D personnel from the labor market to participate in the innovation process. The precise and efficient talent matching mechanism helps improve innovation efficiency (Ma et al., 2024). At the same time, enterprises can develop online learning platforms to enhance the knowledge accumulation efficiency of R&D personnel, significantly improving their overall labor quality. On the other hand, the construction of digital sharing platforms enhances high-tech industrial chain's perception of the external environment. This increases the accessibility of scientific research resources, and the reduction in non-productive costs allows more funds to be invested in the actual R&D of new products and technologies, effectively promoting the accumulation, integration, and absorption of innovation resources (Zou et al. 2022). More importantly, it helps high-tech industrial chain identify short-term market innovation opportunities and its own development shortcomings, clarifying future innovation needs and achieving optimal input combinations of innovation resources (Mishra and Agarwal, 2010), thereby deeply expanding the market size and penetration of high-tech industrial chain. In terms of innovation output, DIC helps high-tech industries accurately identify market demand and product positioning, improving the predictive and coordination capabilities of high-tech industrial chain and promoting product customization (Lee et al., 2024), thereby enhancing the market penetration and circulation matching of innovation outputs. As DIC expands and improves, digital innovation-sharing platforms will connect upstream and downstream enterprises in the high-tech industrial chain to form an innovation network, continuously promoting the integration and reorganization of innovation knowledge and expanding the boundaries of innovation cooperation. The communication efficiency between different innovation entities is significantly improved, greatly reducing the likelihood of “ineffective innovation” and maximizing effective innovation output (Anser et al., 2020). Based on the above analysis, DIC can significantly improve the innovation efficiency of high-tech industrial chain, thereby enhancing their risk resistance and path-recreation capabilities. Therefore, this paper proposes Hypothesis 3:

H3: DIC effectively improves the innovation efficiency of high-tech industrial chain, thereby enhancing HCR.

- 3 At the level of network operation, digital technologies and platform-based mechanisms facilitate the rational allocation and integration of key production factors within the chain. According to the dynamic capability theory, DIC enhances the ability of high-tech industrial enterprises to perceive changes in the industry and environment. Through technological empowerment, data integration, and platform authorization, it drives the digital transformation and upgrading of traditional production factors, fully leveraging the multiplier effect of data and technology on traditional factors, and promoting the resolution of resource mismatches in high-tech industrial chain, thereby achieving rational factor allocation (Zhang et al., 2024). First, in terms of labor factor allocation, DIC promotes the application of digital technologies such as artificial intelligence and automation, directly replacing some routine and low-skilled jobs in high-tech industrial chain and increasing the demand for high-quality talent, thereby optimizing the human capital structure of high-tech industrial chain. At the same time, big data analysis, online recruitment platforms, and other digital tools enhance the transparency of the labor market, accelerating the matching process between enterprises and employees and significantly reducing labor resource waste caused by frictional unemployment (Ma et al., 2024). Second, in terms of capital factor allocation, DIC has given rise to digital financial information platforms such as JD Baitiao and Ant Credit Pay, which not only broaden the financing channels for related enterprises but also reduce their credit costs. Through project screening and real-time monitoring, the development of digital finance effectively alleviates information asymmetry between supply and demand (Gao et al., 2023). The enhanced transparency of information on both direct and indirect financing strengthens enterprises within the chain. It enables more accurate profitability forecasts, a deeper market understanding of corporate strategy, and reduced uncertainty regarding future prospects (Lei et al., 2026). As a result, it facilitates the flow of capital factors toward regions with higher efficiency within the industrial chain. With the deepening application of digital finance, capital factors will gradually flow to the lower end of the value chain of high-tech industrial chain as the marginal cost of the network economy decreases, alleviating the financing difficulties of small and micro enterprises. The effective allocation of capital factors significantly enhances the overall development scale and flexibility of high-tech industrial chain, enabling them to promptly formulate financial service solutions to fill funding gaps in the face of external shocks. The free flow and interconnection of factors help improve high-tech industrial chain's ability to anticipate external uncertainties, enhancing its dynamic adjustment and adaptability (Jiang and Li, 2024). Based on the above analysis, this paper proposes Hypothesis 4:

H4: DIC effectively improves factor allocation level, thereby enhancing HCR.

Finally, we also construct a technology roadmap to present the impact pathways of DIC on HCR in a more intuitive form, as shown in Figure 4.

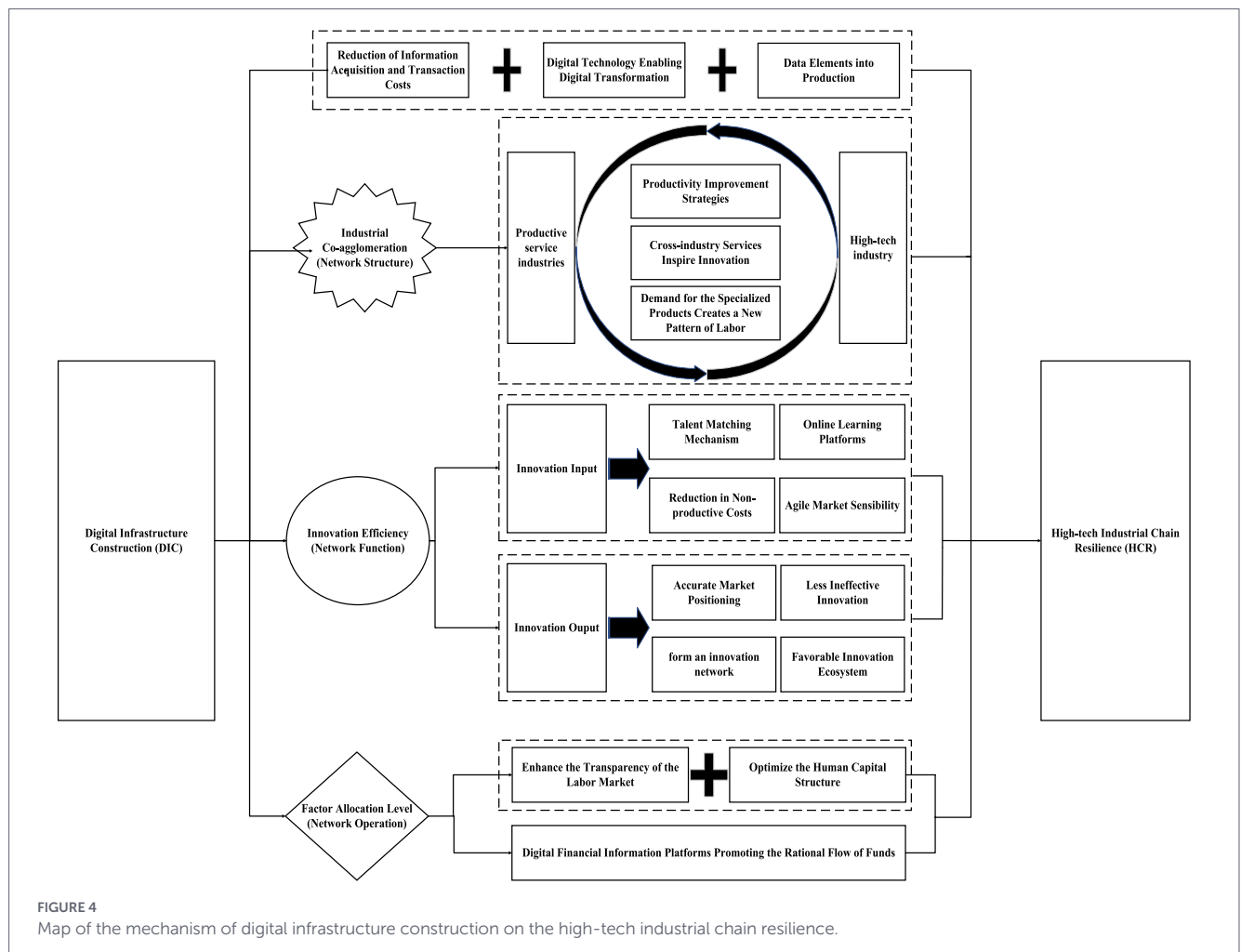
4 Methodology and data sources

4.1 Variable selection

4.1.1 Digital infrastructure construction (DIC)

This paper constructs a comprehensive evaluation index system for DIC from four dimensions based on the definition of “new infrastructure” issued by the *National Development and Reform Commission*: information infrastructure, integrated infrastructure, innovation infrastructure, and government support, selecting 16 secondary indicators (Table 1):

- 1 *Information infrastructure*: as the core component of DIC, information infrastructure plays a crucial role in information dissemination and provides essential hardware infrastructure for various social production and life activities. This paper refers to Chang et al. (2024) and Osei (2024), selecting seven indicators to measure information infrastructure construction: the length of long-distance optical cables, mobile phone switch capacity, internet broadband access port density, mobile phone penetration rate, number of internet domain names, per capita telecommunications business volume, and the number of 5G base stations. Among these, internet broadband access port density is calculated as (number of internet broadband access ports)/(number of internet broadband access users).
- 2 *Integrated infrastructure*: integrated infrastructure reflects the dynamic evolution of the “generational leap” between old and new infrastructure. It not only directly improves the convenience of daily life in terms of “clothing, food, housing, and transportation” but also injects new momentum into the economic development of various industries. Therefore, this paper selects five indicators to measure integrated infrastructure construction: smart highway mileage, smart transportation infrastructure, smart energy infrastructure, smart medical infrastructure, and smart living infrastructure. Among these, “smart highway mileage” is calculated as the sum of the mileage of high-speed railways, highways, first-class roads, and second-class roads in each province. Smart transportation infrastructure, smart energy infrastructure, smart medical infrastructure, and smart living infrastructure are represented by the number of smart transportation enterprises, smart energy enterprises, smart medical enterprises, and smart living enterprises, respectively.
- 3 *Innovation infrastructure*: innovation infrastructure is a key digital infrastructure that promotes the smooth development of various innovation activities, enhances social productivity, and drives high-quality economic development. This paper refers to Yang and Wang (2024), selecting three indicators to measure innovation infrastructure: the number of national-level technology business incubators, the number of university research and development institutions, and the number of national university science parks.
- 4 *Government support*: currently, DIC in China, represented by 5G, artificial intelligence, and industrial internet, is still in its early stages. Such infrastructure requires large upfront



investments, has long economic return cycles, and is heavily influenced by national policies. Therefore, DIC is still government-led at this stage. This paper considers this by manually collecting provincial government work reports from 2008 to 2021 and identifying keywords related to DIC¹. Using Python software for analysis and processing, the ratio of keywords related to DIC to the total vocabulary in each province's government work reports from 2008 to 2021 is calculated as a proxy indicator.

¹ In this paper, the keywords related to *digital infrastructure* are "5G, ICT, digital economy, digital industrialization, digital life, digital technology, data technology, datatization, digital industry, digital infrastructure, data center, data sharing, data security, data service, data governance, data assetization, digital information, digitalization, information technology, information and communication technology, information technology, information industry, information economy, information infrastructure, information service, cloud service, cloud technology, cloud application, cloud platform, cloud computing, cloud, smart manufacturing, intelligent equipment, smart economy, intelligent factory, intellectualized, intelligent, intelligent economy, intelligent science and technology, intelligent technology, intelligent terminal, e-commerce, e-technology, e-government affairs, software, software engineering, software technology, mobile communication, mobile payment, communication infrastructure, communication technology, knowledge economy, big data, artificial intelligence, robotics, blockchain, Internet of Things, industrial Internet, Internet, Internet platforms, modern information networks, networking technology, computer technology, networking science and technology, online."

4.1.2 High-tech industrial chain resilience (HCR)

This paper interprets HCR based on Martin's "resistance-recovery-adaptation-innovation" theoretical framework (Wu, 2023; Wu A. et al., 2023) to construct a comprehensive evaluation index system for HCR from three levels: "resistance and recovery capability," "adjustment and adaptation capability," and "innovation and transformation capability" (Table 2):

- Resistance and recovery capability*: this is the foundational capability of HCR, referring to the ability of the industrial chain to withstand external shocks and disturbances and return to its original or a new state after being impacted. This paper selects indicators from three aspects: (A) Industrial Chain Development Scale: The larger the industrial chain development scale, the more extensive and diversified the enterprises and markets it covers, and the richer its resource and knowledge reserves. Even after being impacted by risks, high-tech industrial chain will only be slightly affected and can recover quickly. Referring to Gu and Zhang (2023), this paper selects the number of high-tech industrial chain enterprises, total profits of high-tech industrial chain, and main business income to describe the development scale of high-tech industrial chain. (B) Industrial Chain Dependence: The stronger the industrial chain dependence, the more likely it is that problems in one segment of the chain will trigger a chain reaction,

TABLE 1 Comprehensive evaluation indicator system of digital infrastructure construction.

Variable	First level indicator	Second level indicator	Indicator unit	Data sources
Digital Infrastructure Construction	Information Infrastructure	The length of long-distance optical cables	10,000 kilometers	<i>China Statistical Yearbook, Statistical Yearbook of the Provinces</i>
		Mobile phone switch capacity	10,000 households	<i>China Statistical Yearbook, Statistical Yearbook of the Provinces</i>
		Internet broadband access port density	Units/households	<i>China Statistical Yearbook, Statistical Yearbook of the Provinces</i>
		Mobile phone penetration rate	/	<i>China Statistical Yearbook, Statistical Yearbook of the Provinces</i>
		The number of 5G base stations	10,000	<i>China Statistical Yearbook, Statistical Yearbook of the Provinces</i>
		Number of internet domain names	10,000	<i>China Statistical Yearbook, Statistical Yearbook of the Provinces</i>
		Per capita telecommunications business volume	10,000 yuan per person	<i>China Statistical Yearbook, Statistical Yearbook of the Provinces</i>
	Integrated Infrastructure	smart highway mileage	10,000 kilometers	<i>China Statistical Yearbook, Statistical Yearbook of the Provinces</i>
		smart transportation infrastructure	Pieces	<i>Qichacha Platform (www.qichacha.com)</i>
		smart energy infrastructure	Pieces	<i>Qichacha Platform (www.qichacha.com)</i>
		smart medical infrastructure	Pieces	<i>Qichacha Platform (www.qichacha.com)</i>
		smart living infrastructure	Pieces	<i>Qichacha Platform (www.qichacha.com)</i>
	Innovation Infrastructure	the number of national-level technology business incubators	Pieces	<i>China Torch Statistical Yearbook</i>
		the number of university R&D (research and development) institutions	Pieces	<i>China Science and Technology Statistical Yearbook</i>
		the number of national university science parks	Pieces	<i>China Torch Statistical Yearbook</i>
	Government Support	the ratio of keywords related to digital infrastructure to the total vocabulary in each province's government work reports	/	Official website of the provincial government

TABLE 2 Comprehensive evaluation indicator system of high-tech industrial chain resilience.

Variable	First level indicator	Second level indicator	Third level indicator	Indicator unit	Data sources
High-tech Industrial Chain Resilience	Resistance and Recovery Capability	Industrial Chain Development Scale	The number of high-tech industrial chain enterprises	Unit	China High Technology Statistical Yearbook
			Total profits of the high-tech industrial chain	Trillion yuan	China High Technology Statistical Yearbook
			Main business income	Trillion yuan	China High Technology Statistical Yearbook
		Industrial Chain Dependence	Foreign direct investment	Trillion yuan	China High Technology Statistical Yearbook
			Total import and export volume of the high-tech industrial chain	Million dollars	China High Technology Statistical Yearbook
		Internal Structure of the Industrial Chain	The proportion of resource-based high-tech industrial chain and	/	China High Technology Statistical Yearbook
			The proportion of technology-based high-tech industrial chain	/	China High Technology Statistical Yearbook
	Adjustment and Adaptation Capability	Industrial Chain Flexibility	The industrial diversification index (HHI, Herfindahl–Hirschman Index)	/	China High Technology Statistical Yearbook
			The average number of employees	Pieces	China High Technology Statistical Yearbook
		Industrial Chain Development Efficiency	Project completion rate	/	China High Technology Statistical Yearbook
			Labor productivity	Million yuan per person	China High Technology Statistical Yearbook
			Average profit rate	/	China High Technology Statistical Yearbook
		Industrial Chain Turnover Capability	Fixed asset investment	Trillion yuan	China High Technology Statistical Yearbook
			Technology introduction and transformation costs	10,000 yuan	China Science and Technology Statistical Yearbook
	Innovation and Transformation Capability	Industrial Chain Transformation Potential	The full-time equivalent of R&D (Research & Development) personnel	Per person-year	China Science and Technology Statistical Yearbook
			Internal and external R&D expenditure	10,000 yuan	China Science and Technology Statistical Yearbook
			The number of R&D institutions with R&D activities	Pieces	China Science and Technology Statistical Yearbook
		Industrial Chain Transformation Capability	The number of patent applications	Pieces	China Science and Technology Statistical Yearbook
			New product development projects	Pieces	China Science and Technology Statistical Yearbook
			Valid invention patents	Pieces	China Science and Technology Statistical Yearbook
		Industrial Chain Innovation Linkages	The innovation degree centrality	/	Calculated by Ucinet
			The innovation closeness centrality	/	Calculated by Ucinet

paralyzing the entire chain and affecting its overall resistance and recovery capabilities. Given the current tense international situation, the role of export trade in driving domestic economic growth is gradually weakening, and enhancing the autonomous control of industrial chain is imperative. This paper selects foreign direct investment and total import and export volume of high-tech industrial chain to represent industrial chain dependence. (C) Internal Structure of the Industrial Chain: A reasonable industrial chain structure facilitates the rapid flow of information, efficient allocation of resources, and flexible scheduling of production processes, while an unreasonable internal structure can lead to excessive resource concentration and operational inefficiencies. Based on Wang and Yang (2005) interpretation of high-tech industrial chain, this paper divides high-tech industrial chain into resource-based high-tech industrial chain and technology-based high-tech industrial chain according to inter-industry linkage factors.² The proportion of resource-based high-tech industrial chain and technology-based high-tech industrial chain is used to reflect the spatial structure of the industrial chain, calculated as (total output value of pharmaceutical manufacturing + information chemical manufacturing)/(total output value of high-tech industries) and (total output value of aviation, aerospace, and equipment manufacturing + electronic and communication equipment manufacturing + computer and office equipment manufacturing + medical equipment and instrument manufacturing)/(total output value of high-tech industries), respectively.

- 2 *Adjustment and adaptation capability*: this is the optimization capability of HCR, referring to the ability of the industrial chain to promptly identify and address deficiencies and quickly adapt to new development environments after being impacted by risks. This paper selects indicators from three aspects: (A) Industrial Chain Flexibility: The speed at which the industrial chain adjusts its structure and finds feasible alternative solutions to cope with market demand fluctuations after being impacted. This paper selects the average number of employees and the industrial diversification index to reflect industrial chain flexibility. The industrial diversification index is calculated based on the HHI (Herfindahl–Hirschman Index) following He et al. (2022). (B) Industrial Chain Development Efficiency: Industrial chain development efficiency is positively correlated with its stress resistance, representing resource allocation optimization capability, collaborative efficiency, and the sustainable competitive

advantage of the industrial chain. This paper selects project completion rate, labor productivity, and average profit rate to calculate industrial chain development efficiency (Zheng, 2023). (C) Industrial Chain Turnover Capability: Excellent industrial chain turnover capability represents flexible resource scheduling capability. It contributes to long-term strategic planning and promotes innovation and cooperation among different participants to enhance the operational efficiency of enterprises on the industrial chain (Chandra and Grabis, 2009). This paper uses fixed asset investment and technology introduction and transformation costs to measure this capability.

- 3 *Innovation and transformation capability*: this is the transformative capability of HCR, referring to the ability of the industrial chain to continuously promote new models, formats, and modes to enhance its competitiveness in the face of changing external factors. This paper selects indicators from three aspects: (A) Industrial Chain Transformation Potential: A good innovation foundation is a prerequisite for various innovation activities, which is the key to creating a favorable innovation atmosphere and enhancing innovation capabilities. This paper selects the full-time equivalent of R&D (Research & Development) personnel, internal and external R&D expenditure, and the number of R&D institutions with R&D activities to describe the transformation potential of high-tech industrial chain (Yang and Liu, 2024). (B) Industrial Chain Transformation Capability: The more innovation outputs the industrial chain produces, the higher the likelihood of breaking “path dependence,” and the stronger the industrial chain transformation capability. Referring to Liu X. et al. (2024), this paper selects the number of patent applications, new product development projects, and valid invention patents. (C) Industrial Chain Innovation Linkages: This refers to the geographical spatial innovation linkages consisting of dynamic flow of innovation entities and factors across different regions. Following Ye and Qian (2021), this paper uses an *enhanced gravity model and social network analysis* model to calculate the innovation degree centrality and innovation closeness centrality to reflect the innovation linkages of high-tech industrial chain.³

Finally, this paper uses the comprehensive evaluation index system method to measure the level of DIC and HCR. To ensure the scientific accuracy of the measurement results, it combines the entropy method and the CRITIC method for weighting. The entropy method focuses on the degree of dispersion of indicators, while the CRITIC method emphasizes the contrast intensity and conflict of indicators.

Figures 5, 6 respectively illustrate the changes in DIC and HCR from 2008 to 2021. The overall level of DIC exhibited a fluctuating upward trend within a narrow range, rising from 0.2610 (the mean value from 2008 to 2021) in 2008 to 0.2745 in 2021. Eleven regions registered levels above the mean, with the overall development pattern presenting a situation of “high in the east and low in the west,” and the sample distribution displaying an “olive-shaped” structure. The overall resilience level of high-tech industrial chain exhibited a continuous upward trend, increasing from 0.2561 in 2008 to 0.2681 in 2021.

2 *Resource-based high-tech industrial chain* refers to the high-tech industrial chain in which the enterprises in the chain have strong technical content, but the upstream and downstream relationships are determined by resources. For example, the Chinese medicine industrial chain includes the process of “R&D and standard-setting of traditional Chinese medicine - high-quality breeding of seed sources - processing of finished traditional Chinese medicine - extraction of drugs - industrialization of traditional Chinese medicine.” *Technology-based high-tech industrial chain* refers to the high-tech industrial chain in which each enterprise in the chain is determined by technical factors. For example, the electronic information industrial chain includes the process of “network technology research and development - network equipment - transmission network - access network equipment - network integration.” It contains information technology, communication technology, Internet technology and microelectronics technology.

3 See Appendix A - Supplementary Material for specific calculation process.

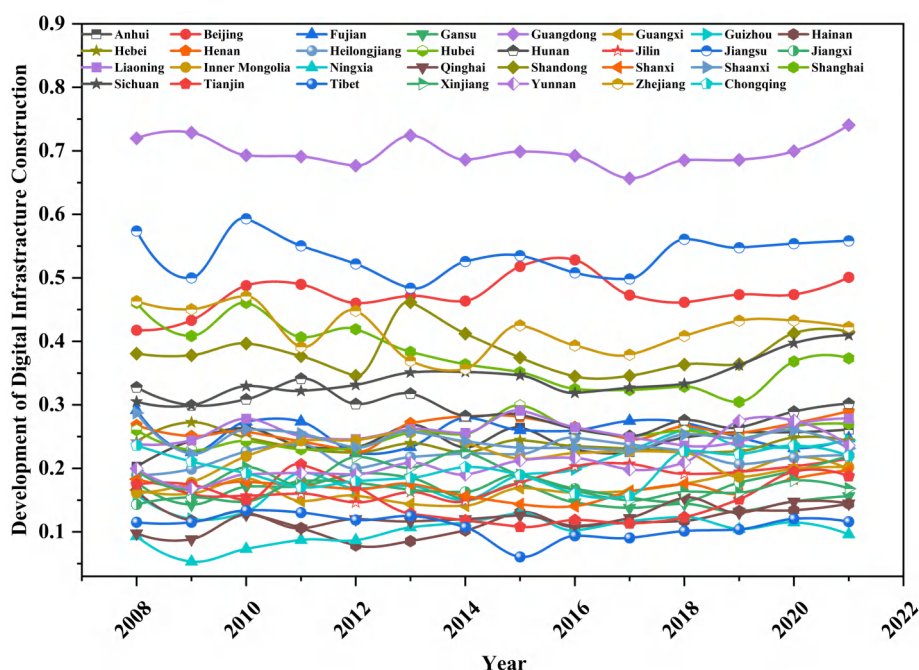


FIGURE 5
Comprehensive assessment scores of digital infrastructure construction in different provinces.

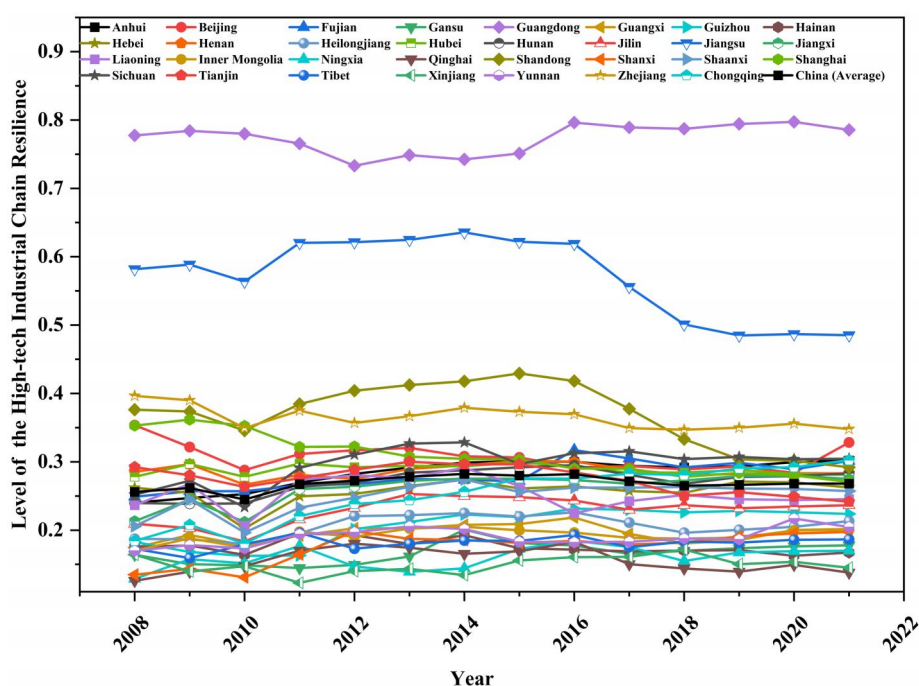


FIGURE 6
Comprehensive assessment scores of high-tech industrial chain resilience in different provinces.

Thirteen regions registered levels above the mean, predominantly located in coastal areas, with the sample distribution approximating a “cone-shaped” structure.

4.1.3 Control variables

To eliminate the impact of omitted variables on the theoretical analysis results, this paper selects the following six control variables to include in the model: (1) Government Intervention (GOV): Local

governments generally promote the development of high-tech industrial chain through policy support, park construction, market regulation, and other measures, playing a crucial role in HCR. This paper uses the ratio of local fiscal general budget expenditure to GDP to measure this. (2) Technology Support (TECH): Direct financial support for science and technology helps promote the agglomeration and development efficiency of high-tech industrial chain. This paper uses the ratio of local fiscal science and technology expenditure to GDP to measure this. (3) Average Education Level (EDU): High-tech

industries are knowledge- and technology-intensive industries. Differences in education levels affect people's understanding of high-tech industries and their ability to acquire related skills, leading to regional differences in innovation capabilities to impact HCR. This paper uses the logarithm of the total number of people with a college degree above 6 years old to measure this. (4) Regional Entrepreneurship Vitality (ENTER): Strong entrepreneurship vitality promotes innovation and integration among upstream, midstream, and downstream segments of high-tech industrial chain, which is crucial for enhancing the innovation and transformation capabilities of HCR. This paper uses the logarithm of the number of urban private enterprises and self-employed individuals to measure this. (5) Foreign Dependence (FDI): Appropriate openness to foreign investment promotes the flexibility of high-tech industrial chain, but excessive reliance on foreign investment may weaken the resistance and recovery capabilities of high-tech industrial chain in the face of external risks. This paper uses the logarithm of the total investment of foreign-invested enterprises to measure this. (6) Market Consumption Scale (MAR): A larger consumption scale represents strong regional consumption and production capabilities, attracting more industrial development resources, which is important for promoting the development scale and efficiency of high-tech industrial chain. This paper uses the ratio of total retail sales of consumer goods to GDP to measure this.

4.1.4 Mediation variables

- 1 *Industrial co-agglomeration (SIA)*: this paper refers to the *Statistical Classification of Productive Services (2019)* issued by the National Bureau of Statistics and selects seven productive service industries closely related to high-tech industry for calculation: real estate, finance, education, leasing and business services, transportation, warehousing and postal services, scientific research and technical services and geological exploration, and information transmission and computer services and software. The industrial co-agglomeration index is calculated as follows (Zheng and He, 2022):
 - A Calculate the industrial agglomeration level (location quotient) of high-tech industries (Equation 1) and productive service industries (Equation 2):

$$SIAP_{i,t} = \frac{E_{iH}/E_H}{E_i/E} \quad (1)$$

$$SIAP_{i,t} = \frac{E_{iP}/E_P}{E_i/E} \quad (2)$$

Where $SIAH_{i,t}$ and $SIAP_{i,t}$ represent the industrial agglomeration levels of high-tech industries and productive service industries, respectively. E_{iH} and E_{iP} represent the number of employees in high-tech industries and productive service industries in region i , respectively. E_H and E_P represent the total number of employees in high-tech industries and productive service industries nationwide, respectively. E_i represents the total number of employees in region i , and E represents the total number of employees

nationwide. For convenience, the calculation is based on urban unit employment.

- B Calculate the industrial co-agglomeration index (SIA) (Equation 3) based on the improved E-G index:

$$SIA_{i,t} = 1 - \frac{|SIAP_{i,t} - SIAH_{i,t}|}{SIAP_{i,t} + SIAH_{i,t}} + (SIAP_{i,t} + SIAH_{i,t}) \quad (3)$$

2 Innovation Efficiency (IE)

- A Construct a three-stage DEA model evaluation index system for high-tech industrial chain innovation efficiency (Lin et al., 2021) (Table 3).
- B Use the three-stage DEA model to measure high-tech industrial chain innovation efficiency:

① Stage 1: Use the super-efficiency SBM model as the basic model of the three-stage DEA to measure initial innovation efficiency (Equation 4) (Du et al., 2010).

$$\min \rho^* = \frac{\frac{1}{m} \sum_{i=1}^m \left(\frac{\bar{x}}{x_{ik}} \right)}{\frac{1}{s} \sum_{s=1}^s \left(\frac{\bar{y}}{y_{sk}} \right)}, \quad s.t. \left\{ \begin{array}{l} \bar{x} \geq \sum_{j=1, j \neq k}^n x_{ij} \lambda_j, i=1, \dots, m \\ \bar{y} \leq \sum_{j=1, j \neq k}^n y_{ij} \lambda_j, s=1, \dots, r \\ \bar{x} \geq x_k \\ \bar{y} \leq y_k \\ \lambda_j > 0 \\ j=1, \dots, n \end{array} \right. \quad (4)$$

Where, ρ^* denotes the efficiency value of the decision unit; n denotes the total number of decision units; m , r are the number of inputs and outputs of the decision unit, respectively; $\bar{x}$, $\bar{y}$ are the slack variables of the inputs and outputs of the decision unit, respectively; x_{ij} and y_{ij} are the actual inputs and outputs values of the decision unit, respectively, and λ_j is the corresponding weight.

② Stage 2: Construct an SFA model to analyze the causes of efficiency loss and adjust the input variables of other decision-making units based on the most efficient decision-making unit (Equations 5–6).

$$s_{ik} = f^i(z_k; \beta^i) + \delta_{ik} + \theta_{ik}, i=1, \dots, m; k=1, \dots, n \quad (5)$$

$$\hat{x}_{ik} = x_{ik} - \left[\max(z_k \hat{\beta}^i) - z_k \hat{\beta}^i \right] + \left[\max(\hat{\delta}_{ik}) - \hat{\delta}_{ik} \right] \quad (6)$$

Where, s_{ik} is the difference between the i^{th} input variable of the k^{th} decision unit and the most efficient target input. $f^i(z_k; \beta^i)$ is the specific effect of the environmental variable on the input variable. z_k is the environmental variable, and β^i is the estimated coefficient of the environmental factor. $\delta_{ik} + \theta_{ik}$ denotes the combined error term, δ_{ik} is the management inefficiency error term, and θ_{ik} is the random error term, which are independent and uncorrelated. $\hat{x}_{ik}$ denotes adjusted input variables.

③ Stage 3: Replace the original input variables with the adjusted input variables $\hat{x}_{ik}$ and recalculate using the super-efficiency SBM model to obtain the final innovation efficiency (IE).

TABLE 3 Evaluation index system of three-stage DEA model for innovation efficiency of high-tech industrial chain.

Variable	Variable layer	Variable name	Indicator unit	Data sources
Innovation Efficiency of the High-tech Industrial Chain	Input Layer	Internal R&D expenditure	Trillion yuan	China High Technology Statistical Yearbook
		The number of R&D personnel	Person	China High Technology Statistical Yearbook
		Costs of technology introduction, acquisition, digestion and adaptation	Trillion yuan	China High Technology Statistical Yearbook
		The number of R&D institutions	Pieces	China High Technology Statistical Yearbook
	Output Layer	Valid invention patents	Pieces	China High Technology Statistical Yearbook
		Revenue from sales of new products	Trillion yuan	China High Technology Statistical Yearbook
	Environmental Variable	Level of economic development (GDP per capita)	10,000 yuan	China Provincial Statistical Yearbook
		Labor quality (number of students in general higher education)	10,000 person	China Provincial Statistical Yearbook
		Degree of market competition (number of industrial enterprises above designated size)	Pieces	China Provincial Statistical Yearbook

3 Factor Allocation Level (FAE):

- A Calculate the capital mismatch index φ_{Ki} and the labor mismatch index φ_{Li} :

$$\varphi_{Ki} = \frac{1}{1 + \tau_{Ki}}, \varphi_{Li} = \frac{1}{1 + \tau_{Li}} \quad (7)$$

Where φ_{Ki} and φ_{Li} are the absolute factor price distortion coefficients, which are replaced by the relative price distortion coefficients in the actual measurement process. They are calculated as follows:

$$\hat{\varphi}_{Ki} = \left(\frac{K_i}{K} \right) / \left(\frac{s_i \beta_{Ki}}{\beta_K} \right), \hat{\varphi}_{Li} = \left(\frac{L_i}{L} \right) / \left(\frac{s_i \beta_{Li}}{\beta_L} \right) \quad (8)$$

Where s_i is the real share of economic output of region i , (K_i/K) is the real proportion of capital used in region i to the total amount of capital. (L_i/L) is the real proportion of labor used in region i to the total amount of labor. β_{Ki} and β_{Li} denote the output elasticities of

capital and labor for region i , respectively. $\beta_K = \sum_{i=1}^N s_i \beta_{Ki}$ and $\beta_L = \sum_{i=1}^N s_i \beta_{Li}$ denote the output-weighted factor contribution values

of capital and labor, respectively. Equation 8 reflects the degree of capital and labor factor resource mismatch of region i . If the ratio is greater than 1 ($\tau < 0$), it indicates over-allocation of factors in the region; conversely, if the ratio is less than 1 ($\tau > 0$), it indicates under-allocation of factor resources in the region. In order to avoid the inconsistency of the sign direction of the regression interference, this paper takes the absolute value of τ . The larger the absolute value is, the more serious factor mismatch is.

- B Estimate the output elasticity of capital and labor factors in each region: β_K and β_L .

From Equations 7, 8, it can be seen that the premise of calculating the capital mismatch index τ_{Ki} and the labor mismatch index τ_{Li} is to derive the output elasticity of capital and labor factors, β_K and β_L , which is specifically measured by the *Solow residual method*. Assuming that the production function is a C-D production function with constant returns to scale (Equation 9), the specific form is as follows:

$$Y_{it} = A K_{it}^{\beta_K} L_{it}^{1-\beta_K} \quad (9)$$

Where Y_{it} is the real economic output of province i in period t , measured by provincial GDP. L_{it} is the labor factor stock, expressed as the annual average number of employed people in each province. K_{it} is the capital factor stock, expressed as the fixed capital stock in each province. The source of data is the *China Statistical Yearbook* and the statistical yearbooks of each province. In this paper, the perpetual inventory method is used to calculate it (Equation 10):

$$K_t = I_t / P_t + (1 - \sigma_t) K_{t-1} \quad (10)$$

Where K_t refers to the fixed capital stock in the current period (this paper takes 2008 as the base period), I_t is the nominal gross fixed capital formation in the current period, P_t is the price index of investment in fixed assets. σ_t indicates the depreciation rate, taken as 9.6%, and K_{t-1} represents the fixed capital stock in the previous period.

- C Based on the output elasticities of capital and labor factors, β_K and β_L , calculated in (B), the capital mismatch index τ_{Ki} and the labor mismatch index τ_{Li} are derived, which in turn leads to the

TABLE 4 Descriptive statistics of all variables.

Variable	(1)	(2)	(3)	(4)	(5)	(6)
	Variable symbol	N	Mean	Sd.	Min	Max
High-tech Industrial Chain Resilience	HCR	434	0.269	0.126	0.123	0.797
Digital Infrastructure Construction	DIC	434	0.257	0.134	0.0532	0.741
Government Intervention	GOV	434	0.283	0.200	0.0998	1.354
Technology Support	TECH	434	0.005	0.003	0.002	0.014
Average Education Level	EDU	434	8.255	1.264	3.689	12.16
Regional Entrepreneurship Vitality	ENTER	434	6.366	1.014	3.384	8.712
Foreign Dependence	FDI	434	11.06	1.609	6.280	15.33
Market Consumption Scale	MAR	434	0.390	0.0630	0.220	0.610
Industrial Co-agglomeration Index	SIA	434	2.297	0.770	1.008	5.358
Innovation Efficiency	IE	434	0.218	0.273	0	1
Factor Allocation Level	FAE	434	0.329	0.234	0.0157	1.673

mediating variable FAE (factor allocation level) (Equation 11) in this paper:

$$FAE_i = \frac{1}{2}(\tau_{Ki} + \tau_{Li}) \quad (11)$$

4.2 Data sources

The data used in this paper in the process of empirical analysis come from China Statistical Yearbook, China High Technology Statistical Yearbook, China Science and Technology Statistical Yearbook, and the statistical yearbooks of each province in the past years. Some of the missing values are made up by interpolation⁴. The descriptive statistics of each variable are analyzed in Table 4.

4.3 Methods and models

4.3.1 Benchmark regression model

Based on the analysis of the impact mechanism of DIC on HCR in Section 3, this subsection sets the following benchmark regression model for empirically analyzing the direct impact of DIC on HCR:

$$HCR_{i,t} = \alpha_0 + \beta_0 DIC_{i,t} + \varphi_0 \sum XCV_{i,t} + \mu_i + \tau_t + \varepsilon_{i,t} \quad (12)$$

Where (Equation 12), $HCR_{i,t}$ represents the level of HCR in the region, $DIC_{i,t}$ represents the status of DIC in the region, and $XCV_{i,t}$ represents the control variables affecting the level of HCR. i and t represent the region and the year, respectively. α_0 stand for the intercept term. β_0 represents the degree of the impact of DIC on HCR. φ_0 represents the coefficients of a series of control variables. μ_i is the city fixed effects, controlling differences between regions; τ_t is the time fixed effects, controlling differences due to changes in the macro-environment. $\varepsilon_{i,t}$ is the random interference term.

4.3.2 Mediation effect model

In order to further test the indirect mechanism of DIC on HCR, this paper constructs a mediation effect model on the basis of the benchmark regression model to conduct a profound analysis of DIC on HCR:

$$HCR_{i,t} = \alpha_0 + \beta_0 DIC_{i,t} + \varphi_0 \sum XCV_{i,t} + \mu_i + \tau_t + \varepsilon_{i,t} \quad (13)$$

$$M_{i,t} = \alpha_1 + \beta_1 DIC_{i,t} + \varphi_1 \sum XCV_{i,t} + \mu_i + \tau_t + \varepsilon_{i,t} \quad (14)$$

$$HCR_{i,t} = \alpha_2 + \beta_2 DIC_{i,t} + \xi M_{i,t} + \varphi_2 \sum XCV_{i,t} + \mu_i + \tau_t + \varepsilon_{i,t} \quad (15)$$

Where, $M_{i,t}$ represents the mediating variables. Based on the theoretical analysis in Section 3, this subsection will conduct an in-depth analysis from the three ways of industrial co-agglomeration effect, innovation efficiency enhancement effect and factor allocation level rationalization effect generated by DIC. α_0 , α_1 , α_2 in Equations 13–15 represents the intercept term. β_1 is the strength of the role of different mediation variables on HCR, and φ_1 represents the coefficients of a series of control variables. In Equations 14,15, the direct effect is β_2 , indicating the degree of direct influence of DIC on HCR. $\beta_1 * \xi$ denotes the indirect effect, indicating the degree of the role of each intermediary variable in the process of DIC affecting HCR. The rest of the letters have the same meaning as the benchmark regression model.

⁴ When employing panel data in the analysis, adjacent observations tend to exhibit strong correlation, and the interpolation method can effectively prevent the decline in estimation efficiency. Specifically, the location and pattern of missing values are first identified, distinguishing between isolated missing data and continuous missing data. For isolated missing values, linear interpolation is performed using the two adjacent valid observations; for continuous missing values, linear estimation is conducted based on the valid data points at both ends of the missing sequence. Missing values at the beginning or end of a sequence are not interpolated. Finally, a robustness test is conducted on the interpolated data.

TABLE 5 Results of Hausman test.

Variable	(1)	(2)	(3)	(4)
	OLS	OLS	FE	RE
DIC	0.827*** (0.021)	0.794*** (0.032)	0.072* (0.043)	0.198*** (0.041)
GOV		0.065*** (0.021)	−0.022 (0.034)	−0.008 (0.031)
TECH		−3.886*** (1.380)	1.262 (1.019)	1.441 (1.055)
EDU		−0.005 (0.003)	0.021*** (0.007)	0.022*** (0.007)
ENTER		0.000 (0.006)	0.020*** (0.007)	0.027*** (0.007)
FDI		0.015*** (0.004)	0.002 (0.003)	0.004 (0.003)
MAR		0.021 (0.049)	0.020 (0.026)	0.002 (0.027)
Constant	0.057*** (0.006)	0.794*** (0.032)	0.072* (0.043)	0.198*** (0.041)
N	434	434	434	434
R ²	0.780	0.792	0.260	0.792
Yearfix	NO	NO	YES	YES
Cityfix	NO	NO	YES	YES

***, * denote significant at the 1 and 10% levels, respectively; standard errors of clustering are in parentheses.

5 Empirical analyses

5.1 Benchmark regression results

To determine the benchmark regression model, this paper also conducts a Hausman test.⁵ Table 5 shows that the Hausman test value is 52.54 and significantly rejects the null hypothesis at the 1% level, so this paper uses fixed effects for panel regression. Table 6 presents the benchmark regression results for DIC and HCR. Column (1) shows the regression results without any control variables, indicating that the coefficient of DIC is significantly positive at the 5% level. Columns (2)–(7) use two-way fixed effects and gradually add different numbers of control variables. Column (7) includes all control variables, showing that the coefficient of DIC is 0.072 and remains significant at the 10% level, indicating that DIC has a significant positive impact on HCR, with an average effect of 7.2%.

The empirical results in Column (7) demonstrate that, with the continuous improvement of DIC, the number of enterprises connected to the industrial chain network and the degree of connectivity

have steadily increased, enabling more efficient and smoother information transmission among various segments of high-tech industrial chain. The resulting scale expansion effect has further strengthened the synergy capacity and resource allocation level among relevant enterprises. Specifically, DIC has significantly reduced the costs of information acquisition and enterprise transactions, promoted the widespread application of digital technologies, and exerted the empowering role of data as a production factor. DIC has fully manifested the characteristics of network externalities. It forges a progressive process of “*linkage expansion - synergistic integration - functional reinforcement*,” which accelerates the digital transformation of the high-tech industrial chain and effectively enhances its resilience. This finding corroborates the theoretical mechanism analysis presented in Section 3 of this paper. Regarding the control variables, the coefficients of EDU and ENTER are positive and significant at the 10% level, while GOV, TECH, FDI, and MAR are not significant. This suggests that: (1) Higher regional education levels represent more high-quality human capital, which helps improve the development efficiency and innovation level of high-tech industrial chain, thereby enhancing their resilience. (2) More private enterprises indicate higher regional entrepreneurship vitality, which stimulates the development potential of high-tech industrial chain through expanding market size and intense competition, crucial for enhancing the innovation and transformation capabilities of HCR. (3) Although government intervention, including tax incentives and talent introduction policies, optimizes factor allocation in high-tech industrial chain, the negative but insignificant coefficient of GOV suggests that government

⁵ Before conducting the benchmark regression, this paper performs a series of tests to ensure data stability. The correlation analysis results show that all variables have linear correlations but not causal relationships. The VIF (Variance Inflation Factor) values are all less than 10, indicating no multicollinearity issues. The unit root test results show that all variables are first-order stationary, indicating that the time series is stable. The detailed data will be available by contacting the authors directly.

TABLE 6 Benchmark regression results on the impact of digital infrastructure construction on high-tech industrial chain resilience.

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)
DIC	0.091** (0.044)	0.089** (0.044)	0.081* (0.044)	0.080* (0.043)	0.076** (0.043)	0.073* (0.043)	0.072* (0.043)
GOV		−0.020 (0.035)	−0.022 (0.035)	−0.025 (0.034)	−0.024 (0.034)	−0.025 (0.034)	−0.022 (0.034)
TECH			2.075** (0.981)	2.000** (0.967)	1.575 (0.964)	1.500 (0.972)	1.262 (1.019)
EDU				0.025*** (0.007)	0.022*** (0.007)	0.022*** (0.007)	0.021*** (0.007)
ENTER					0.022*** (0.007)	0.021*** (0.007)	0.020*** (0.007)
FDI						0.002 (0.003)	0.002 (0.003)
MAR							0.020 (0.026)
Constant	0.246*** (0.011)	0.252*** (0.016)	0.245*** (0.016)	0.042 (0.061)	−0.071 (0.069)	−0.084 (0.072)	−0.075 (0.074)
N	434	434	434	434	434	434	434
R ²	0.974	0.974	0.974	0.975	0.976	0.976	0.976
Yearfix	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cityfix	Yes	Yes	Yes	Yes	Yes	Yes	Yes

***, **, * denote significant at the 1%, 5 and 10% levels, respectively; standard errors of clustering are in parentheses.

decisions may be short-sighted and lagging, and the “performance-oriented” mechanism may have adverse effects on HCR (Zhang et al., 2023). (4) Foreign dependence (FDI) and market consumption scale (MAR) are not significant but have positive coefficients, indicating that foreign investment helps improve factor allocation level in high-tech industrial chain, addressing development shortcomings, and has potential benefits for moving high-tech industrial chain toward the high end of the value chain. Market size expansion stimulates supply and demand, extending industrial chain nodes and potentially enhancing the risk resistance capability of high-tech industrial chain.

5.2 Robustness tests

5.2.1 Reducing sample size

To improve the accuracy of the research results, this paper also uses the method of reducing the sample size for re-regression: excluding the top and bottom 5% of provinces, the top and bottom 1% of provinces, and the top three provinces in DIC (Guangdong, Jiangsu, and Beijing). Columns (1)–(3) in Table 7 show that the coefficients of HCR are significantly positive at the 5, 1, and 10% levels, respectively, preliminarily verifying the robustness of the core conclusions.

5.2.2 Replacing the dependent variable

The dependent variable, HCR, is calculated using the entropy-CRITIC comprehensive weighting method. Referring to Ndubuisi et al. (2021), this paper replaces the entropy-CRITIC comprehensive weighting method with the entropy-TOPSIS comprehensive

weighting method to recalculate HCR and conducts re-regression. Column (4) in Table 7 shows that the coefficient of DIC slightly decreases to 0.071 and remains significant at the 10% level, proving the robustness of the benchmark regression results.

5.2.3 Replacing the core explanatory variable

Compared with the comprehensive evaluation index system constructed above, the capital stock of digital infrastructure can also largely reflect the actual situation of DIC. Therefore, this paper refers to Wen et al. (2024) and uses the total capital stock of emerging information, modern energy, and modern transportation facilities to re-measure DIC. This paper calculates the capital stock of digital infrastructure (Equation 16) by deflating fixed investment data related to digital infrastructure with 2008 as the base year and converting it into corresponding capital stock using the “perpetual inventory method.” The specific formula is as follows:

$$K_{t+1} = K_t (1 - \sigma) + I_t \quad (16)$$

Where, the subscript t denotes time; K_t denotes the digital infrastructure capital stock in year t ; K_{t+1} denotes the digital infrastructure capital stock in year $(t + 1)$. I_t denotes the amount of investment in year t . σ denotes the depreciation rate, taken as 9.6%. Column (5) in Table 7 shows the regression results after replacing DIC with “digital infrastructure capital stock.” The coefficient of DIC is 0.089 and significantly positive at the 5% level, verifying the accuracy of the benchmark regression results again.

TABLE 7 Results of robustness tests on the impact of digital infrastructure construction on high-tech industrial chain resilience.

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)
DIC	0.092** (0.039)	0.071* (0.043)	0.136*** (0.039)	0.053*** (0.011)	0.092** (0.044)	0.073* (0.043)	0.103** (0.045)
INFRAS						0.012 (0.011)	
GOV	−0.038 (0.031)	−0.027 (0.034)	−0.025 (0.030)	0.006 (0.043)	0.014 (0.035)	−0.020 (0.034)	−0.013 (0.034)
TECH	0.258 (0.926)	1.161 (1.014)	0.787 (0.963)	0.846 (1.063)	0.763 (1.050)	0.941 (1.059)	2.023* (1.064)
EDU	0.018*** (0.007)	0.021*** (0.007)	0.017*** (0.006)	0.021** (0.009)	0.034*** (0.007)	0.020*** (0.007)	0.018** (0.007)
ENTER	0.016** (0.006)	0.019*** (0.007)	0.011* (0.006)	0.018*** (0.007)	0.018*** (0.007)	0.020*** (0.007)	0.010 (0.007)
FDI	0.001 (0.003)	0.002 (0.003)	−0.002 (0.003)	0.002 (0.003)	0.003 (0.003)	0.002 (0.003)	0.005 (0.003)
MAR	0.037 (0.024)	0.021 (0.026)	0.054** (0.023)	−0.008 (0.027)	0.001 (0.027)	0.019 (0.026)	−0.016 (0.027)
Constant	−0.023 (0.067)	−0.072 (0.073)	0.008 (0.064)	−0.562*** (0.124)	−0.317*** (0.076)	−0.182 (0.121)	−0.022 (0.075)
N	434	434	392	434	434	434	378
R ²	0.969	0.976	0.930	0.977	0.984	0.976	0.979
Yearfix	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cityfix	Yes	Yes	Yes	Yes	Yes	Yes	Yes

***, **, * denote significant at the 1%, 5 and 10% levels, respectively; standard errors of clustering are in parentheses.

5.2.4 Excluding the impact of traditional infrastructure construction

Traditional infrastructure construction is the basic support and operational driver of high-tech industrial chain, enhancing communication efficiency between upstream and downstream segments and playing an important role in improving HCR. In the development of high-tech industrial chain, traditional infrastructure mainly includes power, water, and communication infrastructure essential for production, as well as highways, railways, and bridges involved in logistics and transportation. This paper refers to [Chao et al. \(2024\)](#) and selects “per capita public transport vehicle operations (logarithmic transformation)” as a proxy variable for INFRAS (traditional infrastructure construction) to include in the regression. Column (6) in [Table 7](#) shows that the coefficient of DIC is 0.073 and significant at the 10% level, while the coefficient of INFRAS is positive but not significant. These results suggest that traditional infrastructure construction may no longer meet the economic conditions necessary for improving HCR, while DIC has become a new engine for enhancing HCR in China, again proving the robustness of the benchmark regression results.

5.2.5 Excluding all municipalities from the sample

Considering the differences between provinces and municipalities in administrative level, population density, industrial distribution, and other aspects, which may introduce different analytical

dimensions and challenges, especially the economic linkages between municipalities and neighboring provincial cities (e.g., Shanghai and Suzhou in Jiangsu Province, Chongqing and Chengdu in Sichuan Province), this may introduce noise variables and cause significant deviations in the empirical results. Therefore, this paper excludes the four municipalities (Beijing, Shanghai, Tianjin, and Chongqing) from the sample and ensures that all individual units in the research sample are at the “provincial” level. Column (7) in [Table 7](#) shows that the coefficient of DIC is 0.103 and significant at the 5% level, with both the effect size and significance level significantly higher than the benchmark regression results. These results strongly prove that DIC has become a new engine for improving HCR in China, again confirming the robustness of the benchmark regression results.

5.3 Endogeneity test

This paper refers to [Hu et al. \(2023\)](#) and [Wu W., et al. \(2023\)](#) and selects “the first and second lags of DIC” and “the number of post offices per million people in 1984 multiplied by the telephone penetration rate (units per 100 people) in the previous year (logarithmic transformation)” as instrumental variables, represented by “L. DIC,” “L2. DIC,” and “DIC2,” respectively. These instrumental variables meet the requirements of relevance and exogeneity: (1) The DIC index system constructed in this paper focuses on “information,” “integration,” and “innovation” infrastructure, emphasizing information technology. In the early stages of modern information and internet

TABLE 8 Results of heterogeneity test on the impact of digital infrastructure construction on high-tech industrial chain resilience.

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	First	Second	First	Second	First	Second	First	Second
L. DIC	0.4588*** (10.36)							
L2. DIC			0.2149*** (4.34)					
DIC2					0.1121*** (8.53)			
DIC3							0.0985*** (7.32)	
GOV	−0.0660* (−1.68)	−0.0236 (−0.62)	−0.1083** (−2.32)	0.0178 (0.31)	−0.0540 (−1.12)	0.0414 (0.80)	−0.0618 (−1.64)	0.0176 (0.43)
TECH	1.1898 (1.09)	1.7092* (1.66)	2.1319* (1.65)	0.6662 (0.46)	2.2087* (1.86)	0.3687 (0.29)	2.3584** (2.07)	0.3451 (0.28)
EDU	−0.0024 (−0.31)	0.0212*** (2.97)	−0.0069 (−0.75)	0.0236** (2.43)	−0.0058 (−0.60)	0.0215** (2.11)	−0.0058 (−0.72)	0.0217** (2.52)
ENTER	0.0056 (0.77)	0.0188*** (2.78)	0.0025 (0.30)	0.0161* (1.83)	−0.0019 (−0.25)	0.0201** (2.51)	0.0009 (0.11)	0.0190** (2.34)
FDI	0.0023 (0.71)	0.0003 (0.11)	0.0037 (1.02)	−0.0024 (−0.60)	0.0011 (0.31)	−0.0021 (−0.56)	0.0024 (0.67)	−0.0029 (−0.76)
MAR	0.0037 (0.14)	0.0127 (0.51)	0.0091 (0.30)	0.0137 (0.43)	−0.0432 (−1.44)	0.0153 (0.50)	−0.0306 (−1.04)	0.0125 (0.40)
DIC		0.2041** (2.27)		0.6584*** (2.76)		0.6536*** (5.28)		0.7194*** (4.94)
Constant	0.0838 (1.10)	−0.0709 (−0.98)	0.2043** (2.19)	−0.1856 (−1.64)	−0.5741*** (−4.83)	−0.1770* (−1.92)	−0.1337 (−1.52)	−0.1746** (−2.02)
N	403	403	372	372	434	434	434	434
R ²		0.976		0.965		0.964		0.961
Phase I F-value	107.26***		18.85***		72.75***		53.56***	
Anderson-LM		93.918***		20.508***		69.009***		53.247***
Cragg-Donald Wald F		107.263 [16.38]		18.846 [16.38]		72.746 [16.38]		53.561 [16.38]
Yearfix	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cityfix	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

***, **, * denote significant at the 1%, 5 and 10% levels, respectively; standard errors of clustering are in parentheses. ‘[]’ denotes the critical value at the 10% level for the Stock-Yogo if identified test. The Anderson-LM statistic significantly rejects the original hypothesis that the instrumental variables are not identifiable. And the results of both the first-stage F-value statistic and the Cragg-Donald Wald F’s test indicate that the estimation process does not suffer from a weak instrumental variable problem.

development, post offices and fixed telephones were important cornerstones of information networks. Therefore, this paper uses the number of post offices and mobile phone penetration rates from the 1980s as instrumental variables, meeting the relevance requirement. (2) With the continuous transformation of information technology and industrial structure upgrading, the number of post offices and mobile phone penetration rates from the 1980s have minimal impact on HCR now, meeting the exogeneity requirement. The specific regression results are shown in Table 8. Columns (1)–(4) show the regression results for “L. DIC” and “L2. DIC.” Columns (5)–(6) show the regression results for “DIC2.” The results show that the regression coefficients and signs of DIC on HCR are approximately consistent

with the original estimates and remain significantly positive at the 1% level, with increased significance. This indicates that the model estimation and instrumental variable selection are reasonable and effective, and the benchmark regression results remain robust, confirming that DIC has a significant positive impact on HCR.

5.4 Mediation effect empirical results

5.4.1 Industrial co-agglomeration

Column (1) in Table 9 shows that the regression coefficient of DIC on industrial co-agglomeration is 1.137 and significantly positive at the

TABLE 9 Results of mediation effect on the impact of digital infrastructure construction on high-tech industrial chain resilience.

Variable	(1)	(2)	(3)	(4)	(5)	(6)
	Industrial co-agglomeration		Innovation efficiency		Factor allocation level	
	SIA	HCR	IE	HCR	FAE	HCR
DIC	1.137**	0.039	0.031*	0.070*	−0.435***	0.045*
	(0.529)	(0.040)	(0.168)	(0.041)	(0.142)	(0.043)
SIA		0.030***				
		(0.004)				
IE				0.079***		
				(0.012)		
FAE						−0.063***
						(0.015)
GOV	−0.592	−0.005	0.031	−0.025	0.135	−0.014
	(0.419)	(0.032)	(0.133)	(0.032)	(0.112)	(0.033)
TECH	−1.086	1.295	18.791***	−0.223	−15.795***	0.265
	(12.540)	(0.950)	(3.977)	(0.999)	(3.364)	(1.026)
EDU	0.030	0.020***	−0.026	0.023***	0.002	0.021***
	(0.089)	(0.007)	(0.028)	(0.007)	(0.024)	(0.007)
ENTER	0.000	0.020***	0.022	0.018***	−0.036	0.018***
	(0.084)	(0.006)	(0.027)	(0.007)	(0.023)	(0.007)
FDI	0.009	0.001	−0.045***	0.005*	0.032***	0.004
	(0.039)	(0.003)	(0.012)	(0.003)	(0.010)	(0.003)
MAR	0.991***	−0.009	0.345***	−0.007	0.342***	0.042
	(0.321)	(0.025)	(0.102)	(0.025)	(0.086)	(0.026)
Constant	1.387*	−0.098	0.538**	−0.099	0.216	−0.043
	(0.822)	(0.063)	(0.261)	(0.064)	(0.221)	(0.066)
SOBEL Z	5.914***		10.524***		2.467**	
N	434	434	434	434	434	434
R ²	0.075	0.358	0.335	0.330	0.158	0.292
Yearfix	Yes	Yes	Yes	Yes	Yes	Yes
Cityfix	Yes	Yes	Yes	Yes	Yes	Yes

***, **, * denote significant at the 1%, 5 and 10% levels, respectively; standard errors of clustering are in parentheses.

5% level. With the continuous improvement of DIC, the number of enterprises connected to the high-tech network has gradually increased, and the degree of interconnectedness and synergy among relevant industries has steadily improved. DIC significantly reduces enterprise production and communication costs, while digital services and intelligent management gradually penetrate the production and operation of high-tech industries, increasing the demand for jobs in productive service industries. Leveraging platform-based carriers to realize collaborative operations, it effectively fosters synergistic cooperation and agglomeration effects between high-tech industry and producer services. It further promotes the deep integration and synergistic development of the upstream and downstream segments of the high-tech industrial chain. By reshaping the network structure of the high-tech industrial chain, this process contributes to the enhancement of its resilience. Column (2) shows the regression results after including both DIC and SIA in the model. Only SIA is significantly positive at the 1% level,

and the coefficient of DIC decreases compared to the benchmark regression coefficient ($0.039 < 0.072$) but is not significant. The mediation effect is 0.03, accounting for 47.38% of the total effect. These results prove that industrial co-agglomeration plays a fully mediating role in the impact of DIC on HCR. After controlling for SIA, the direct effect of DIC on HCR disappears, and only the transmission path of “DIC → industrial co-agglomeration → improvement of HCR” exists. In high-tech industrial chain with high knowledge and technology density and high added value, industrial co-agglomeration is particularly important due to rapid technological updates and large market demand fluctuations. Industrial co-agglomeration enhances technical exchanges and resource sharing among related enterprises, improving production efficiency and innovation capabilities, and significantly enhancing the adaptability and innovation transformation capabilities of industrial chain, thereby improving its resilience. The mediation effect of industrial co-agglomeration is established, and H2 is supported.

5.4.2 Innovation efficiency

Column (3) in Table 9 shows that the regression coefficient of DIC on innovation efficiency is 0.031 and significantly positive at the 10% level. This result indicates that DIC significantly reduces labor costs and optimizes resource input efficiency through information transmission and data sharing. As network connectivity expands, innovation factors flow and match more efficiently among enterprises. This accelerates technological R&D, market application, and cross-industry and cross-regional collaborative innovation. Consequently, it helps reduce redundant investment in innovation resources and alleviates resource misallocation. By improving both the output rate and the conversion efficiency of innovation outputs, this process promotes the enhancement of innovation efficiency. Column (4) shows the regression results after including both DIC and IE in the model. Both are significantly positive at the 10 and 1% levels, respectively, and the coefficient of DIC slightly decreases compared to the benchmark regression coefficient ($0.070 < 0.072$). The mediation effect is 0.031, accounting for 3.4% of the total effect. These results prove that innovation efficiency plays a partially mediating role in the impact of DIC on HCR. Although the mediation effect is relatively limited, for high-tech industries with high knowledge density and rapid iteration speed, the enhancement of pivotal functional efficiency strengthens the chain's capacity for adaptation and adjustment to uncertain shocks. The transmission path of "DIC → innovation efficiency improvement → improvement of HCR" still exists. The mediation effect of innovation efficiency is established, and H3 is supported.

5.4.3 Factor allocation level

Column (5) in Table 9 shows that the regression coefficient of DIC on factor allocation level is -0.435 and significantly negative at the 1% level. This indicates that, through digital platforms such as the Internet, cloud computing, and big data technologies, DIC has led to a gradual increase in the number of connected entities, making the allocation pathways for labor and capital factors among different nodes increasingly diversified. The circulation and operational efficiency of resources within the industrial chain have been significantly improved, while the phenomena of misallocation and idleness have been effectively reduced. In this process, data, as a new production factor, participates in social production process, helping enterprises quickly identify market demand, optimize production processes, and enhance innovation capabilities, significantly improving factor allocation level. Column (6) shows the regression results after including both DIC and FAE in the model. Both are significant at the 10 and 1% levels, respectively. DIC is significantly positive, and its coefficient decreases compared to the benchmark regression coefficient ($0.045 < 0.072$). FAE is significantly negative, and the calculated mediation effect is 0.027, accounting for 38.06% of the total effect. These results prove that factor allocation level plays a partially mediating role in the impact of DIC on HCR. The improvement of resource allocation capacity can significantly enhance its capacity for dynamic adjustment and adaptation, thus improving HCR. The transmission path of "DIC → factor allocation level improvement → improvement of HCR" exists. The mediation effect of factor allocation level is established, and H4 is supported.

5.5 Heterogeneity tests

The preceding empirical results have confirmed the promoting effect of DIC on HCR and revealed the underlying mechanism.

However, as illustrated in Figures 5, 6, the levels of both DIC and HCR still exhibit significant regional disparities. To further analyze the potential nonlinear relationship between the two, this subsection conducts two sets of heterogeneity tests: (1) The three major regions of East, Central, and West. Differences in altitude can affect the signal propagation and path loss of DIC, thereby maybe exerting differential impacts on HCR. (2) The Southern and Northern regions. Since the *14th Five-Year Plan*, issues of north–south economic divergence have gradually become prominent due to the continuous southward migration of population and the driving effects of urban agglomeration policies.

Columns (1) to (3) of Table 10 present the heterogeneity results for the three major regions: East, Central, and West, respectively. Among these, the regression coefficient for eastern region is 0.024, significantly positive at the 10% level; the coefficient for central region is 0.037 but not significant; and the coefficient for western region is -0.084 , also not significant. This indicates that the impact of DIC on HCR exhibits marked differences across the eastern, central, and western regions. DIC is more likely to exert a facilitating effect on enterprise synergy and resource allocation within high-tech industrial chain, thereby enhancing its operational stability. Although the central region possesses a certain capacity to undertake high-tech industries, deficiencies remain in the depth of digital technology embedding and the allocation of innovation resources, preventing the full manifestation of the promoting effect. In contrast, constrained by factors such as geographic conditions, population distribution, and industrial foundations in the western region, digital infrastructure investment has difficulty generating an impact on high-tech industrial chain in the short term (Zhang et al., 2025).

Columns (4) and (5) present the heterogeneity analysis results for the southern and northern regions, respectively. The regression coefficient for the south is 0.139, significant at the 5% level; the coefficient for the north is -0.127 , also significant at the 5% level, revealing a significant divergence. In recent years, the southern region has experienced a continuous influx of population, coupled with advantages such as favorable urban agglomeration policies and the concentration of innovation resources, resulting in a relatively high density of high-tech enterprises. Consequently, DIC is more conducive to enhancing R&D synergy efficiency and information transmission speed, thereby strengthening HCR. In contrast, influenced by factors such as population outflow and a relatively low-end industrial structure in the northern region, the degree of high-tech industries agglomeration is comparatively insufficient, and the role of DIC in supporting innovation synergy and resource integration remains limited.

6 Conclusions and policy implications

6.1 Conclusion

This paper uses provincial panel data from 2008 to 2021 to establish evaluation index systems for DIC and HCR, and conducts empirical analysis using panel regression models and mediation effect models. The main conclusions are as follows: First, the benchmark regression coefficient of DIC is significantly positive at the 10% level, indicating that DIC promotes HCR, with an average effect of 7.2%. The experimental conclusions remain valid after a series of endogeneity and robustness tests. Second, in addition to the direct effect, DIC

TABLE 10 Results of heterogeneity tests.

Variable	(1)	(2)	(3)	(4)	(5)
DIC	0.024*	0.037	−0.084	0.139**	−0.127**
	(0.075)	(0.087)	(0.057)	(0.062)	(0.060)
GOV	0.255**	−0.046	−0.059*	−0.343***	0.040
	(0.118)	(0.159)	(0.030)	(0.125)	(0.034)
TECH	0.779	−0.596	2.684	4.581***	−1.057
	(2.103)	(1.794)	(1.704)	(1.506)	(1.419)
EDU	0.016	0.016	0.003	0.043***	0.004
	(0.018)	(0.014)	(0.007)	(0.012)	(0.008)
ENTER	0.033**	−0.027**	0.027***	0.048***	−0.003
	(0.013)	(0.012)	(0.008)	(0.011)	(0.008)
FDI	−0.002	−0.004	−0.000	0.000	−0.009
	(0.005)	(0.007)	(0.004)	(0.004)	(0.005)
MAR	−0.115**	0.083	0.090**	−0.021	0.061*
	(0.046)	(0.059)	(0.038)	(0.068)	(0.031)
Constant	−0.018	0.318*	0.014	−0.360***	0.300***
	(0.170)	(0.189)	(0.073)	(0.130)	(0.097)
N	182	84	168	210	224
R ²	0.981	0.951	0.923	0.984	0.936
Yearfix	Yes	Yes	Yes	Yes	Yes
Cityfix	Yes	Yes	Yes	Yes	Yes

***, **, * denote significant at the 1%, 5 and 10% levels, respectively; standard errors of clustering are in parentheses.

also promotes HCR through three mediation paths: industrial co-agglomeration, innovation efficiency, and factor allocation level. Among these, industrial co-agglomeration exhibits a fully mediating effect, while innovation efficiency and factor allocation level exhibit partially mediating effects. Third, the heterogeneity tests indicate that the promoting effect of DIC on HCR exhibits significant regional disparities. This effect is more pronounced in the eastern and southern regions, which enjoy stronger economic vitality. In contrast, it is relatively weak, or even suppressive, in the central, western, and northern regions, where the industrial foundation is comparatively weaker.

6.2 Policy implications

Compared with traditional theories that primarily focus on the direct output effects brought about by DIC, the empirical findings of this paper suggest that the role of DIC in empowering the enhancement of HCR is manifested across three dimensions: structure, function, and operation, with its impact mechanism exhibiting distinct networked and systemic characteristics. Therefore, a key theoretical contribution of this paper lies in revealing the intrinsic mechanism through which DIC acts on HCR via multi-dimensional pathways. This is examined from the perspective of network externalities and the industrial chain network. It thereby provides a clearer analytical framework for understanding the role of infrastructure in the context of the digital economy. Based on the above findings, this paper proposes the following policy recommendations.

- 1 At the level of industrial co-agglomeration, greater emphasis ought to be placed on integrating DIC with the actual industrial layout, with priority assigned to promoting co-agglomeration between high-tech industry and producer services. On the one hand, the economies of scale generated by DIC should be fully leveraged to promote interfacing between high-tech industry and producer services such as R&D, logistics, and finance. Improving the data circulation system and enhancing network connectivity can elevate the level of information interconnection and business synergy across industries. This forms cross-industry correlation networks with a higher degree of interconnectedness, thereby facilitating the optimal enhancement of HCR. On the other hand, a simplistic, undifferentiated sprawl-based construction model should be avoided. Instead, DIC should be carried out in an orderly manner tailored to regional development characteristics, which strengthens the “agglomeration-diffusion” effects of core regions. Specifically, the core strategy of “*the East leading and the Central and Western Regions Energizing*” should be followed, while key integrated computing projects such as the “*National Computing Network to Synergize East and West*” initiative should be supported to promote the rational layout of DIC. Mechanisms for data sharing and platform interfacing should also be refined to drive the balanced improvement of regional industrial foundation capacity.

- 2 From the perspective of innovation efficiency, DIC should move from being “scale-oriented” to “function-oriented,” with particular emphasis on enhancing the capacity of digital sharing platforms to lower costs and facilitate innovation conversion. On the one hand, through open innovation platforms, DIC can reduce the cost for high-tech enterprises to acquire technological information and innovation resources, thereby decreasing redundant and ineffective investments. At the same time, the driving role of chain-leading enterprises generates demonstration effects, accelerating the commercialization, application, and diffusion of technological achievements throughout the entire industrial chain. However, this process depends heavily on institutional development and the platform environment. Small and medium-sized enterprises often suffer from insufficient digital application capacity and outdated collaboration channels, which may cause platform construction to become merely formalistic, leading to significant gaps between input and actual output. Therefore, policies should focus on deepening the market-oriented reform of data factors and building cross-level collaborative innovation resource platforms. By guiding more high-tech enterprises to deeply embed themselves into the innovation system of the industrial chain network, for instance, through “industry-university-research” synergy pattern, the innovation efficiency of the high-tech industrial chain can be improved to enhance its resilience.
- 3 In terms of factor allocation level, priority should be given to strengthening the ability of DIC to match labor, capital, and data within the high-tech industrial chain. Although the continuous improvement of DIC has significantly reduced information asymmetry and alleviated factor misallocation in the high-tech industrial chain, regions with a weak factor foundation still struggle to improve factor allocation through digital investment in the short term. Therefore, blind technological investment should be avoided. Instead, labor market information platforms and digital financial systems should be established in line with the industrial foundation and market demand, so as to broaden formal financing channels for high-tech enterprises. Furthermore, the fundamental role of data factors should be strengthened by refining data trading and circulation mechanisms. At the same time, mechanisms for risk monitoring, financial supervision, and data emergency management should be put in place. This will enable various factors to participate more effectively in the production process and enhance the high-tech industrial chain's capacity to anticipate changes in the external environment.

Data availability statement

The original contributions presented in the study are included in the article/[Supplementary material](#), further inquiries can be directed to the corresponding author.

Author contributions

XZ: Conceptualization, Supervision, Writing – review & editing. SL: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing.

Funding

The author(s) declared that financial support was received for this work and/or its publication. This study was supported by the Beijing Municipal Social Science Foundation (grant number: Z19112) and the Pyramid Talent Engineering Training Program Project of Beijing University of Civil Engineering and Architecture (grant number: JDJQ20200305).

Conflict of interest

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Supplementary material

The Supplementary material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/frsus.2026.1767126/full#supplementary-material>

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